

Can you design a route with no wrong turn?

Learning intention

Explain how heart structure supports double circulation and calculate cardiac output from supplied data.

Mission: follow blood through two linked circuits.

A correct route needs both destination and direction.

Direction, exchange and rate

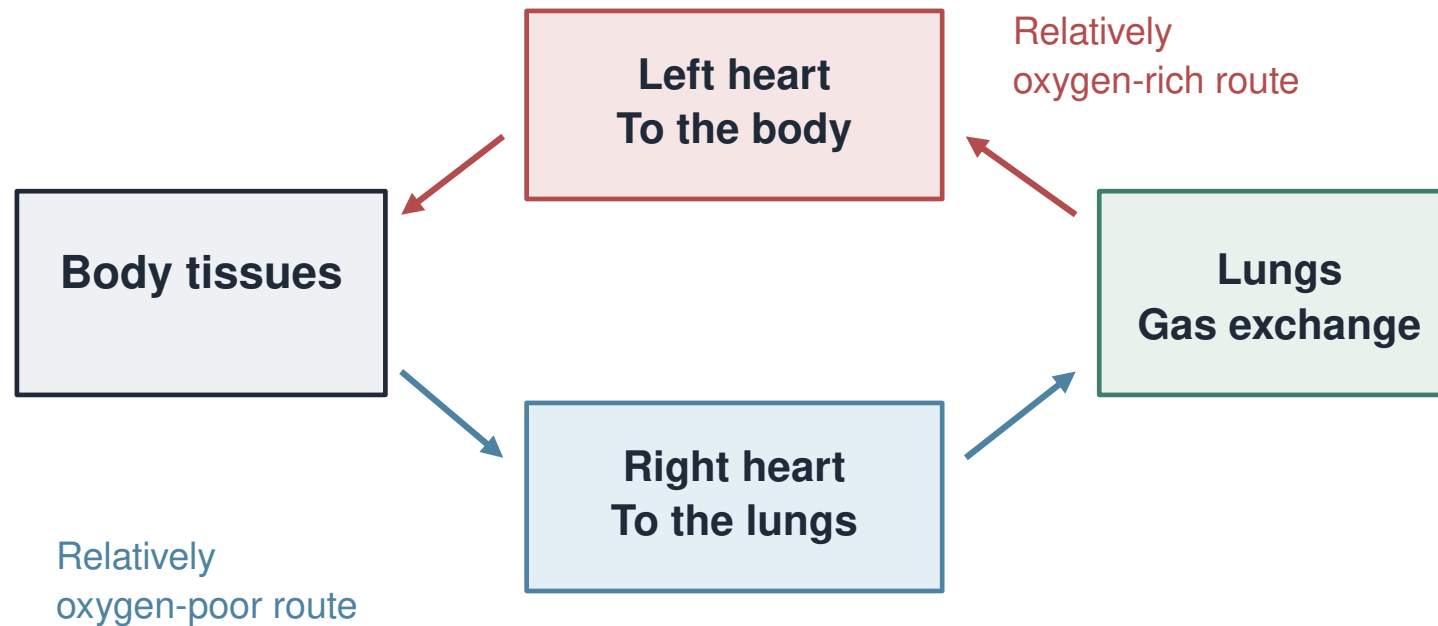
R1. Where does blood gain oxygen from air?

R2. What does a per-minute rate include?

R3. Does a larger pressure mean the same thing as a larger volume?

Trace the full double circuit

Two linked circuits, one continuous route



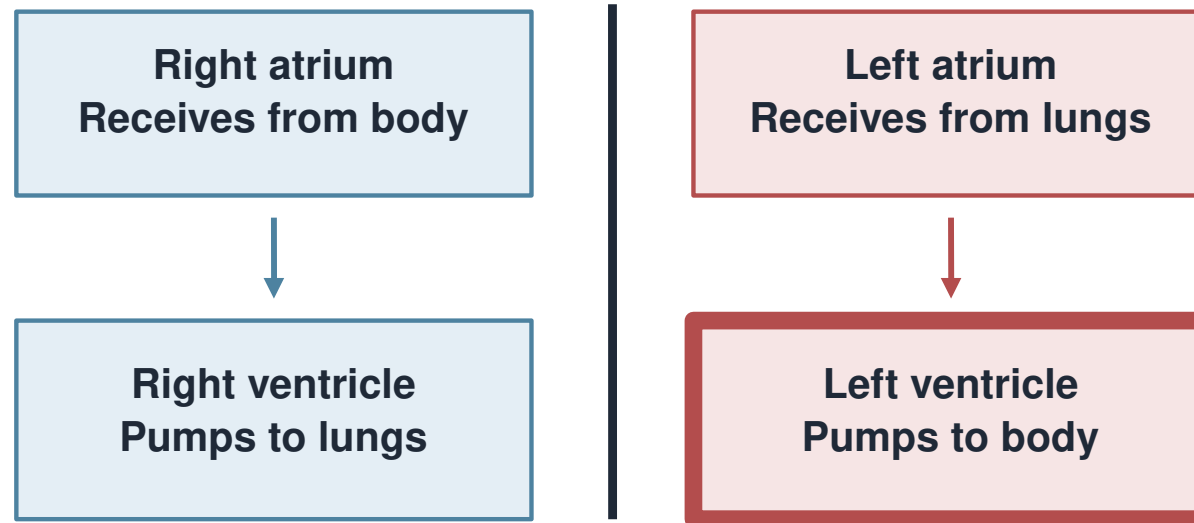
Artery = away from heart; vein = towards heart.

Body → right heart → lungs → left heart → body. Arteries go away; veins return towards the heart.

Explain structure using its job

Atria receive; ventricles pump out

Septum separates sides



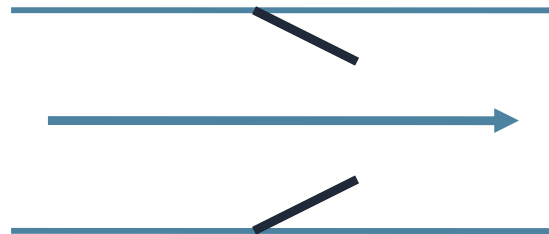
Body's right is shown left; this is a structure model.

Atria receive; ventricles pump out. The septum separates the two sides. The left ventricle generates higher systemic pressure.

Separate the pump from the valve

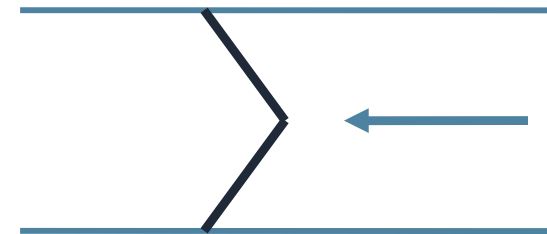
A valve permits forward flow and limits backflow

Forward pressure: open



Flow passes through
when the valve opens

Reverse pressure: closed



Leaflets close against
a reverse pressure

Heart muscle creates pressure; valves direct the flow.

Muscle contraction changes pressure. Valves open or close as pressure differences change. They help prevent backflow.

Route engineering desk

W1. Match the four named vessels to their routes. Add the missing atrium→ventricle steps.

Journey segment	Major vessel
Body → right atrium	Vena cava
Right ventricle → lungs	Pulmonary artery
Lungs → left atrium	Pulmonary veins
Left ventricle → body	Aorta

Schematic adult double-circulation route.

Pump data challenge

W2. Calculate output for P and Q. Check both heart rate and stroke volume.

Record	Beats/min	mL/beat
P	80	55
Q	100	70

Constructed pump records; not pupil measurements.

Repair the wall claim

W3. “The thicker left ventricle proves it always pumps more blood than the right.”

Replace volume with the relevant mechanism.

Which definition survives the lung route?

Choose the accurate statement.

- A** The pulmonary artery is an artery because it carries blood away from the heart.
- B** Every vein must carry relatively oxygen-poor blood.
- C** Valves supply the force that pumps blood.

Produce a route-and-pump brief

Choose your response route.

Show the route, a structural reason and one calculation.

Make every arrow explain something

Name the vessel and direction.

Keep pressure, volume and rate distinct.

Audience checks the exception

L1. Share one vessel claim.

Audience asks: “Would it still work for the pulmonary vessels?”

A route and a calculation need units

A correct name tells us where blood goes.

A correct unit tells us what the number measures.

Three useful checks

E1. Which chamber pumps blood to the lungs?

E2. Why is the left-ventricle wall thicker?

E3. Calculate $60 \text{ mL/beat} \times 90 \text{ beats/min}$.

Next useful step

Vessel direction • chamber route • pressure • output units.